

MARIUM YOUSUF

Curriculum Vitae

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RESEARCH VISION

Develop robust data-driven frameworks for understanding the complex dynamics of the brain, with a particular emphasis on modeling functional and effective connectivity in neural systems.

EDUCATION

Ph.D. in Applied Mathematics (Defended June 17, 2026)	Aug 2026
M.S. in Applied Mathematics	Aug 2023
M.S. in Computer Science	May 2022
University of Arizona, Tucson AZ	
B.S. in Mathematical Sciences , <i>summa cum laude</i>	Dec 2017
Northern Illinois University, DeKalb IL	
Associates in Science	May 2015
McHenry County College, Crystal Lake IL	

ACADEMIC APPOINTMENTS

Ph.D. Student , University of Arizona (UArizona)	
Research	
<i>Department of Mathematics</i>	Fall 2022-Summer 2026
Developing a probabilistic graphical model to extract neural effective connectivity from spike trains.	
Keywords: Replay, Brain Connectivity, Causal Discovery, Network Inference, Stochastic Modeling	
Advisors: Jean-Marc Fellous, Michael Chertkov	
<i>Department of Computer Science</i>	Fall 2019-Spring 2022
Implemented an approximation of G-Wishart marginal likelihood to learn sparse graphical structures representing different levels of functional brain connectivity.	
Primary Instructor	
Elements of Calculus, Dept. of Mathematics	Summer 2024, Summer 2025
College Algebra, Dept. of Mathematics	Spring 2024, Spring 2025
Calculus Preparation, Dept. of Mathematics	Fall 2023
Teaching Assistant	
Understanding Data, Dept. of Mathematics	Spring 2023
College Algebra, Dept. of Mathematics	Fall 2022
Discrete Data Structures, Dept. of Computer Science	Fall 2019, Yr. of 2021
Graduate Research Aide , Argonne National Laboratory	Summers 2021-2023
Automated high-throughput TEM and X-ray mouse brain image processing by integrating visualization systems (WebKnossos, NeuroGlancer) and reducing reliance on manual tools like TrakEM2.	
Graduate Research Assistant , Missouri University of Science and Technology	
<i>Department of Computer Science</i>	Fall 2018-Summer 2019
Pre-processed and analyzed data collected from dementia patients to infer the role of sedentary body movements in early diagnosis of dementia.	
Pre-Doctoral Intern , Argonne National Laboratory	Apr-Aug 2018
Research Aide Built Python tools for efficient visualization of real-time data from sensors located in Chicago for an Array of Things project.	
Lecturer Prepared Python and Jupyter Notebook materials and lectured in a three-day Big Data Visualization Camp for rising high-school seniors.	
Undergraduate Research Aide , Argonne National Laboratory	Summer 2017
Configured Apache Spark in Jupyter Notebook to analyze real-time simulated data for visualization tasks.	
Undergraduate Teaching Assistant , Northern Illinois University (NIU)	
UNIX and Networking, Dept. of Computer Science	Fall 2017

MANUSCRIPTS

Yousuf, M., Pagnier, L., Fellous, J.M., Chertkov, M. 2026. *Inferring Directed Functional Connectivity Graphs from Neural Spike Trains*. Manuscript under review at Neural Computation. March 2026

CONFERENCE CONTRIBUTIONS

Yousuf, M., Pagnier, L., Chertkov, M., & Fellous, J.-M. (2026). *Causality in replay: Detecting effective connectivity from spike trains*. Poster presented at Dynamics Days, Tucson, AZ, Jan 9–11.

Yousuf, M., Pagnier, L., Chertkov, M., & Fellous, J.-M. (2025). *Causality in replay: Detecting effective connectivity from spike trains*. Poster presented at Neuroscience 2025 (Society for Neuroscience), San Diego, CA, Nov 15–19.

Yousuf, M., Pagnier, L., Chertkov, M., & Fellous, J.-M. (2025). *Causality in replay: Comparing methods to detect effective connectivity from spike trains*. Poster presented at NITMB MathBio Convergence Conference, Chicago, IL, Aug 11–18.

Yousuf, M., Chertkov, M., & Fellous, J.-M. (2024). *Hippocampal replay and sleep's hidden language: Methods for detecting functional connectivity from spike trains*. Poster presented at Neuroscience 2024 (Society for Neuroscience), Chicago, IL, Oct 5–9.

Yousuf, M., Chertkov, M., & Fellous, J.M. (2023). *Detecting replay in multi-unit spiking data: Bayesian networks*. Poster presented at Neuroscience 2023 (Society for Neuroscience), Washington, DC, Nov 11–15.

INVITED AND LOCAL PRESENTATIONS

Investigating the success of multiple-choice assessments in a math classroom. Invited presentations at CIRTl Network Teaching-as-Research Presentations (virtual), Apr 8, 2026; and at College Teaching ShareFest (virtual), Apr 17, 2026.

Investigating the success of multiple-choice assessments in a math classroom. Poster presented at the Graduate Interdisciplinary Programs Student Research Showcase, University of Arizona, Tucson, AZ, Dec 11, 2025.

Causality in Hippocampal Replay: Detecting Functional Connectivity from Spike Trains. Invited presentation at the Human Augmented Analytics Group, Georgia Institute of Technology (virtual), Apr 15, 2025.

Hippocampal replay and sleep's hidden language: Methods for detecting functional connectivity from spike trains. Contributed talk at the Arizona Women's Symposium in Mathematics, Flagstaff, AZ, Sep 21, 2024.

Detecting replay in multi-unit spiking data: Bayesian networks. Invited poster presentation at the Graduate Interdisciplinary Programs Student Research Showcase, University of Arizona, Tucson, AZ, Dec 6, 2024.

Detecting replay in multi-unit spiking data: Bayesian networks. Poster presented at the Arizona Women's Symposium in Mathematics, Flagstaff, AZ, Nov 19, 2023.

Diagnosing Dementia Using Smart Chair, Wearable Device, and Testing Apps. Presentation at the Annual Graduate Research Symposium, Intelligent Systems Center, Rolla, MO, Apr 3, 2019.

ADDITIONAL ACADEMIC TRAINING

UArizona NSF Research Training Group Fall 2024, Fall 2025
Funded through NSF-supported research group focused on modern computational methods for data-driven modeling and applications.

Simons Laufer Mathematical Sciences Institute Summer 2025
Selected to attend a summer graduate workshop on Local Limits of Random Graphs held at Université Paris-Saclay Mathematics Institute in France.

Center for the Integration of Research, Teaching, and Learning (CIRTl) Fall 2021–Fall 2025
Completed all designations (*Associate, Practitioner, & Scholar*) in the CIRTl's three-tiered teaching certificate program, with training in evidence-based undergraduate STEM teaching.

AWARDS, HONORS, AND SCHOLARSHIPS

Trainee Professional Development Award (1000 USD), Society for Neuroscience	Nov 2025
Grogan Scholarship Award (6000 USD), Dept. of Mathematics, UArizona	Fall 2025
Herbert E. Carter Travel Award (600 USD, 100 USD), Graduate College UArizona	2024, 2023
TA of the Month, Dept. of Computer Science, UArizona	Oct 2021
Grace Hopper Student Scholar	Oct 2019
Norma K. Stelford Mathematics Endowment, NIU	Dec 2017
(graduating senior in mathematics with the highest GPA)	
The Clarence Ethel Hardgrove Mathematics Scholarship, NIU	2015–2016
(incoming transfer with excellent prior record in mathematics)	
International Undergraduate Scholarship, NIU	2015–2017

PROFESSIONAL DEVELOPMENT

Mentor, STAR Lab, UArizona 2024–2026

Mentored high school seniors in conducting data-driven scientific research	
<i>Chapter Officer</i> , SIAM UArizona Chapter	2024-2026
Served as a Treasurer and Vice President	
Secured 490 USD funds from the SIAM board and assisted in planning chapter events	
Co-organized and built website for SIAM mini-conference for graduate students across disciplines	
<i>Mentor</i> , Human Augmented Analytics Group, Georgia Tech	Spring 2025
Co-advised a Master's researcher on a computer vision project	
<i>Panelist</i> , Graduate Teaching Assistants' Orientation and Training, UArizona	Aug 2024
Invited to serve in GTA training for incoming graduate students in the Dept. of Mathematics	
<i>Mentor</i> , Undergraduate Mathematical Modeling, UArizona	Spring 2024
Mentored an undergraduate team for a capstone project on learning languages using Markov Chains	
<i>Volunteer</i> , Outreach Program BASIS Oro Valley High School, Oro Valley AZ	Mar 2023, 2024
Brain- and memory-inspired educational activities for 6th-graders	
<i>Mentee</i> , Women in STEM Mentorship Program, UArizona	2020-2021
<i>Developer</i> , Healthcare Hackathon 2019 by Codefi, Cape Girardeu MO	July 2019
<i>Developer</i> , AI Chess Tournament by Sandia National Laboratory, Rolla MO	May 2019
<i>Developer</i> , PickHacks Hackathon by Major League Hacking, Rolla, MO.	2018
<i>Volunteer</i> , Hopper for Grace Hopper Conference	2018